NAME – CHIRAG

ROLL - 261410202983

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SEMESTER - 1

Q1.

Environmental Studies is a multi-disciplinary area of study because it draws from multiple disciplines for information. Environmental issues involve nature, humanity, sociology, and technology, thus cannot be addressed entirely through just a single discipline. Different disciplines will work together to better understand how to protect our environment. Examples of other disciplines that make up the multi-disciplinary course of study called Environmental Studies include (but are not limited to):

1. **Biology** - Assists in understanding life forms (plants and animals) and the ecosystems they exist in as well as the relationship between biodiversity. When an example of this is the clearing of forests; this will help to look at how cutting down the forest area affects the wildlife.

2. **Chemistry** - Helps in the study of various types of pollution (air, water and soil), as well as how chemicals from factories pollute rivers and lakes.

3. **Physics** - Assists in the understanding of energy, climate, and natural processes. An example is that both solar and wind energy are examples of how physics has helped the world with respect to environmentally friendly ways to produce energy.

4. **Geography** - Assists in gaining information about the earth (landforms, rivers and mountains), weather and natural resources. Geography also aids with understanding the causes and effects of natural disasters such as floods and droughts.

5. **Economics** - Provides the principles as to how to use natural resources in an efficient manner. Economics can assist both governments and businesses by providing them with the means to create appropriate policies to ensure that there are fair balances between economic growth and protection of the environment.

6. **Sociology and Political Science** - These subjects aid with understanding how humans behave, the environmental laws and government policies which are used to manage human interaction with the environment.

Q2.

Ecological succession is a natural process that takes place over time, whereby one type of plant and animal community replaces another type of such communities. Ecological succession can occur when an area that has been disrupted by some force returns to a more stable, balanced, and mature state (i.e., ecological succession occurs when a simple plant-animal community transitions into a more complex, balanced and stable ecosystem).

Ecological succession can take years to establish new ecosystems, and it continues until a mature (climax) ecosystem has been established that will remain in equilibrium with its environment until it is disrupted by another intense disturbance. Ecological succession plays an important role in the evolution and development of ecosystems and contributes to the overall diversity of all biotic organisms in an ecosystem. Ecological succession occurs because of natural processes following disturbances (e.g., flooding, wildfire, volcanic activity) or because of human activity in establishing new types of habitats. Ecological succession is one of the processes by which ecosystems change, and hydrosere is the aquatic facet of this process. The aquatic environment sustains different forms of life, while the terrestrial environment supports a different set of organisms.

Hydrosere begins in a body of water (e.g., pond, lake, marsh) and passes through several specific vegetation stages, as described below:

1. **Phytoplankton** – The very first stage of hydrosere, during which tiny plants and algae are present in the water column.

2. **Submerged plants** – An intermediate stage in hydrosere, during which submerged vegetation (e.g., water lilies) develops below the surface of the water.

3. **Floating plants** – The next stage, wherein several species grow from just below to above the water's surface.

4. **Reed-sedge swamp** – Typically characterized by tall grass and sedge-like vegetation growing in shallow water.

5. **Grass** – The grass stage is the first time that land vegetation appears after the last water-loving organisms (e.g., rush, cattail) disappear.

6. **Forest** – The final stage of hydrosere (climax community); it's possible for all land vegetation to be fully established and the ecological community to be at stability.

Xerosere, the xerophytic (dry) counterpart of hydrosere, begins with the same stages as hydrosere, but in an extremely dry, barren, or rocky environment (e.g., desert).

1. **Lichens** – The very first stage of xerosere, when lichens can colonize bare rock.

2. **Mosses** – The next stage, during which mosses develop on the same bare rock and, as a result, create the beginnings of soil (moss soil).

3. **Herbs** – The next stage, where low-growing herbaceous plants and grasses begin to occupy the created soil.

4. **Shrubs** – The next stage of xerosere, where shrubs eventually replace the previous population of herbaceous plants.

5. **Forest** – The final stage (climax community) of xerosere, where most of the vegetation will have reached its peak and have formed one of the most stable ecosystems possible.

For example - a rocky site that over the years has gradually transitioned from barren to a developed forest ecosystem.

Q3.

Renewable energy is that which occurs naturally, such as wind, sunshine and rainfall. Natural resources, such as the ocean or trees, are also renewable sources of energy. Therefore, renewable energies do not run out as they are replaced infinitely over time on our planet's surface. Renewable sources of energy have a lesser environmental impact than fossil fuels have. For example, when we burn fossil fuels (e.g., coal, oil and gas), they emit gases into our atmosphere that contribute to global warming and air pollution. With an increase in the world's demand for energy and a finite supply of fossil fuels, renewable sources of energy have become essential for the future. Below are five major renewable sources of energy:

1. **Solar Energy** – Solar energy is produced from sunlight. Solar panels turn sunlight into energy, and this energy can be used in homes and businesses.

2. **Wind Energy** – Wind energy is produced using moving air. Wind turbines capture the power of wind to create electricity and because wind energy is produced from wind and does not cause air pollution.

3. **Hydropower** – Hydropower is created when the energy produced from moving or falling water is converted into electricity. Dams are used for this purpose.

4. **Biomass Energy** – Biomass energy is created from plants, agricultural products and animal waste. Biomass energy may be used to provide heat for cooking, heating, or as a source of electricity, etc.

5. **Geothermal Energy** - Geothermal energy comes from heat that is found beneath the Earth’s surface. This heat energy can be used to create electricity and to provide heat.

The future supply of energy will be largely dependent on renewables as they provide a clean, long-term option that does not deplete. The primary advantages of renewable energy include:

1. Renewable energy resources are naturally available to us and can be utilized indefinitely.

2. Renewable energy resources help lessen pollution and carbon emissions, which are both significant contributors to global warming.

3. Renewable energy resources promote greater energy independence and enhance long-term energy security from imported fossil fuels.

Q4.

The contamination of bodies of water (such as oceans, streams, rivers, lakes, ponds and groundwater) with harmful materials is what is referred to as **water pollution**. Water pollution compromises the ability of water for use in agriculture, drinking or aquatic life. Water pollution results from both human activity and natural events. Generally, the sources of water pollution can be classified as point sources or non-point sources.

1. **Point Source Pollution** - A point source of water pollution refers to an identifiable point at which pollution originates. Point sources of pollution are generally easier to identify and control than are non-point sources. For example, an identified discharge of waste from a factory, sewage from a drain, and treated wastewater from sewage treatment plants.

2. **Non-Point Source Pollution** - Non-point sources of water pollution do not reference an identifiable point or location. Non-point sources of water pollution include, for example, fertilizers and pesticides washed off agricultural fields, rainwater that travels on roads carrying oil, plastic, and garbage; soil erosion from construction activity; and runoff from construction activity.

The following are some of the best examples of how **water pollution affects the environment**.

1. Affects Fish, plants, and other organisms in the water.
2. It promotes some diseases to be spread (example contains cholera, typhoid, and dysentery).
3. Chemicals/toxins kill aquatic life.
4. They'll generate too much growth of water plants via over-fertilizing, resulting in reducing oxygen in the water and thus harming fish.
5. Polluted water will no longer be usable for drinking, agricultural production, or industrial purposes.
6. Polluted water will disrupt the natural balance of the aquatic ecosystem.

Q5.

Environmental laws are created by governments to protect the environment and natural resources. Ecological/Law importance includes the following.

1. Environment Laws control air, water and land pollution.
2. Protects forests, wildlife and natural resources.
3. Established ensuring a healthy environment today and for future generations.
4. Ensures all industries and individuals are held accountable for the protection of the environment.
5. Provides means for sustainable development and protects biodiversity.

The following environmental legislation is very important in India:

1. **The Water (Prevention and Control of Pollution) Act 1974**; this legislation regulates water pollution and aims to restore and maintain water quality of surface waters (i.e., rivers, lakes, and ponds) and prohibits industries from discharging untreated waste into water sources.

2. **The Air (Prevention and Control of Pollution) Act 1981**; this legislation regulates air pollution produced by industrial emissions and from motor vehicles and assists in maintaining clean air; and Air Pollution Control Boards frequently test the air to ensure compliance with the law.

3. **The Environment (Protection) Act 1986**; this legislation was introduced following the Bhopal disaster and provides the central government with the authority to protect the environment and to regulate and control pollution.

4. **The Wildlife Protection Act 1972**; this legislation protects wildlife from hunting and illegal trade and establishes and provides for the administration of National Parks, Wildlife Sanctuaries, and Protected Areas for the preservation of biodiversity.

Q6.

Disaster management is a systematic way of managing how you respond to disasters and how you plan for them to help reduce their impact on people, buildings, and the environment. Activities such as preparation for disasters; responding to disasters; and recovering from disasters all fall under the auspices of disaster management. The primary aim of disaster management is to save lives, reduce damages associated with disasters, and assist communities in their rapid recovery from disasters. Disasters can be natural or created by humans and can result in loss of life and loss of properties and resources.

There are two categories that disasters can fall into as follows:

Natural Disasters - Natural disasters happen when there are unexpected occurrences due to natural forces at work on the earth. Examples are earthquakes, floods, cyclones, droughts, landslides, and tsunamis. These types of disasters cannot be influenced by humans; however, there are planning methodologies that will allow you to effectively respond to the impact of these types of disasters.

Man-Made Disasters - Man-made disasters occur because of human error, human negligence, or human accidents. Examples of man-made disasters are industrial accidents, chemical leaks, nuclear accidents, fire accidents, oil spills, and/or terrorist acts. To help avoid man-made disasters, you can implement safety measures along with awareness programs.

The four phases of disaster management are:

1. **Mitigation**: Mitigation refers to the measures taken to minimize the potential damage from a disaster before it occurs. Some common mitigation activities are - Earthquake-resistant construction, Tree planting for soil erosion control, Building flood control mechanisms. The goal of mitigation is to reduce the amount/duration of damage and losses that can occur because of a disaster.

2. **Preparedness**: Preparedness refers to planning and training that takes place prior to an emergency. Some unique examples of preparedness activities include - Conducting emergency drills, developing disaster response plans, storing emergency supplies, including food, water and medical supplies, providing education to the public on how to prepare for emergencies. Preparedness provides individuals with the knowledge and training to respond quickly and effectively during an emergency.

3. **Response**: Response refers to all actions taken immediately during and after an emergency/disaster. Some examples of response activities include - Search and rescue, first aid/medical treatment, Evacuation of people affected by the disaster, providing food, water and shelter to those who have evacuated their homes. The primary goal of the response phase is to save lives and lessen human suffering caused by the disaster.

4. **Recovery**: Recovery refers to the process of returning to pre-disaster conditions after a disaster has occurred. Some examples of recovery activities include - Rebuilding homes, roads, schools, etc., Restoring/repairing potable water and electrical services, Providing financial/emotional assistance to disaster victims. The recovery phase may be a lengthy process taking weeks, months, or even a couple of years depending on how severe the disaster was.